Numerical Analysis homework VI

LI YIJIAN 3230102477 *

(Mathematics and Applied Mathematics 2302), Zhejiang University

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I. Detailed proof of Thm

(a)

By Newton's interpolation

x_i				
-1	$y(-1)$			
0	$y(0)$	$y(0)-y(-1)$		
0	$y(0)$	$y'(0)$	$y'(0)-y(0)+y(-1)$	
1	$y(1)$	$y(1)-y(0)$	$y(1)-y(0)-y'(0)$	$-y'(0)+(y(1)-y(-1))/2$

Then $p_3(t) = y(-1) + [y(0)-y(-1)](t+1) + [y'(0)-y(0)+y(1)](t+1)t - y'(0)t^2(t+1)$.
Then Becasue the interval are symmetric about 0; So the integral of the odd function are zero. So:

$$\int_{-1}^{1} p_3(t)dt = \int_{-1}^{1} y(-1) + y(0) - y(-1) + [-y(0) + \frac{y(1)+y(-1)}{2}]t^2 dt = \frac{1}{3}(y(-1)+4y(0)+y(1))$$

Which is the same as the result by Simpson's rule.

(b)

By (a) we know $y(t) - p_3(t) = \frac{y^{(4)}(\zeta(t))}{4!}t^2(t^2-1)$. So:

$$E^S(y) = \int_{-1}^{1} \frac{y^{(4)}(\zeta(t))}{4!}t^2(t^2-1)dt \xlongequal{\text{by Mean Value}} \frac{y^{(4)}(\zeta(t))}{4!}\int_{-1}^{1} t^2(t^2-1)dt = -\frac{y^{(4)}(\zeta)}{90}$$

(c)

Let $f \in C^4[a,b]$ and let $n = 2m$ be an even integer. Set

$$h = \frac{b-a}{n}, \qquad x_j = a + jh \quad (j = 0, 1, \ldots, n).$$

Partition $[a,b]$ into m panels $[x_{2k}, x_{2k+2}]$ $(k = 0, 1, \ldots, m-1)$.

Fix k and apply the result of parts (a)–(b) on the reference interval $[-1,1]$ by the change of variables

$$x = x_{2k+1} + ht, \qquad t \in [-1,1], \qquad dx = h\,dt,$$

and define $y(t) = f(x_{2k+1} + ht)$. Then

$$\int_{x_{2k}}^{x_{2k+2}} f(x)\,dx = h\int_{-1}^{1} y(t)\,dt = h\left(\frac{1}{3}\big(y(-1)+4y(0)+y(1)\big) - \frac{1}{90}y^{(4)}(\xi_k)\right)$$

*Electronic address: 3230102477@zju.edu.cn

for some $\xi_k \in (-1, 1)$. Noting that $y(-1) = f(x_{2k})$, $y(0) = f(x_{2k+1})$, $y(1) = f(x_{2k+2})$, and

$$y^{(4)}(t) = h^4 f^{(4)}(x_{2k+1} + ht),$$

we obtain

$$\int_{x_{2k}}^{x_{2k+2}} f(x)\,dx = \frac{h}{3}\Big(f(x_{2k}) + 4f(x_{2k+1}) + f(x_{2k+2})\Big) - \frac{h^5}{90} f^{(4)}(\eta_k),$$

where $\eta_k = x_{2k+1} + h\xi_k \in (x_{2k}, x_{2k+2})$.

Summing over $k = 0, 1, \ldots, m-1$ yields the composite Simpson's rule:

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[f(x_0) + 4\sum_{\substack{j=1 \\ j \text{ odd}}}^{n-1} f(x_j) + 2\sum_{\substack{j=2 \\ j \text{ even}}}^{n-2} f(x_j) + f(x_n)\right] + E_n,$$

with the error term

$$E_n = -\frac{h^5}{90}\sum_{k=0}^{m-1} f^{(4)}(\eta_k).$$

Therefore,

$$|E_n| \le \frac{h^5}{90}\, m \max_{x\in[a,b]} |f^{(4)}(x)| = \frac{b-a}{180}\, h^4 \max_{x\in[a,b]} |f^{(4)}(x)|.$$

Equivalently, there exists $\xi \in (a, b)$ such that

$$E_n = -\frac{b-a}{180}\, h^4 f^{(4)}(\xi).$$

II. Approximate the number of the subintervals

Let $f(x) = e^{-x^2}$, $a = 0$, $b = 1$, and $h = \frac{b-a}{n} = \frac{1}{n}$.

(a) Composite trapezoidal rule

The error bound for the composite trapezoidal rule is

$$|E_n^T| = \frac{b-a}{12}\, h^2 |f''(\xi)| \le \frac{b-a}{12}\, h^2 \max_{x\in[a,b]} |f''(x)|$$

Compute

$$f''(x) = (4x^2 - 2)e^{-x^2}.$$

On $[0, 1]$, $|f''(x)| \le 2$, and in fact $|f''(0)| = 2$, hence

$$|E_n^T| \le \frac{1}{12} \cdot \frac{1}{n^2} \cdot 2 = \frac{1}{6n^2}.$$

Imposing $\frac{1}{6n^2} < 5 \times 10^{-7}$ gives

$$n^2 > \frac{1}{3 \times 10^{-6}} \qquad \Rightarrow \qquad n > 577.35.$$

Thus

$$\boxed{n = 578}.$$

(b) Composite Simpson's rule

For n even, the composite Simpson error bound is

$$|E_S| = \frac{b-a}{180}\, h^4 |f^{(4)}(\xi)| \le \frac{b-a}{180}\, h^4 \max_{x\in[a,b]} |f^{(4)}(x)|$$

Compute

$$f^{(4)}(x) = (16x^4 - 48x^2 + 12)e^{-x^2}.$$

On $[0,1]$, $|f^{(4)}(x)| \le 12$, and $|f^{(4)}(0)| = 12$, hence

$$|E_S| \le \frac{1}{180} \cdot \frac{1}{n^4} \cdot 12 = \frac{1}{15n^4}.$$

Imposing $\frac{1}{15n^4} < 5 \times 10^{-7}$ gives

$$n^4 > \frac{1}{7.5 \times 10^{-6}} \qquad \Rightarrow \qquad n > 19.1.$$

Since

$$\boxed{n = 20}.$$

III. Gauss-Laguerre quadrature formula

(a)

$$\pi_2(t) = t^2 + at + b$$

is orthogonal to $\mathbb{P}_1$,suffices to impose orthogonality against the basis $1, t$:

$$\int_0^\infty \pi_2(t)e^{-t}\,dt = 0, \qquad \int_0^\infty t\,\pi_2(t)e^{-t}\,dt = 0.$$

Using the hint $\int_0^\infty t^m e^{-t}\,dt = m!$, we get

$$\int_0^\infty (t^2 + at + b)e^{-t}\,dt = 2! + a\,1! + b\,0! = 2 + a + b = 0,$$

and

$$\int_0^\infty t(t^2 + at + b)e^{-t}\,dt = \int_0^\infty (t^3 + at^2 + bt)e^{-t}\,dt = 3! + a\,2! + b\,1! = 6 + 2a + b = 0.$$

hence $a = -4$, $b = 2$. Therefore

$$\boxed{\pi_2(t) = t^2 - 4t + 2}.$$

(b)

The nodes are the zeros of π_2:

$$x_{1,2} = 2 \mp \sqrt{2}.$$

The corresponding weights are

$$w_1 = \frac{2+\sqrt{2}}{4}, \qquad w_2 = \frac{2-\sqrt{2}}{4}.$$

Hence the 2-point Gauss–Laguerre quadrature formula is

$$\int_0^\infty e^{-t} f(t)\,dt = w_1 f(2-\sqrt{2}) + w_2 f(2+\sqrt{2}) + E_2(f).$$

By Hermite interpolation at x_1, x_1, x_2, x_2,we know

$$f(t) - H_3(t) = \frac{f^{(4)}(\xi_t)}{4!}\,(t-x_1)^2(t-x_2)^2.$$

Let $I(f) = \int_0^\infty f(t)e^{-t}\,dt$ and $Q_2(f) = w_1 f(x_1) + w_2 f(x_2)$. Since H_3 is a cubic polynomial and the 2-point Gauss–Laguerre rule is exact for all polynomials of degree ≤ 3, we have

$$\int_0^\infty H_3(t)e^{-t}\,dt = Q_2(H_3).$$

But $H_3(x_i) = f(x_i)$ for $i = 1, 2$, hence

$$Q_2(H_3) = w_1 H_3(x_1) + w_2 H_3(x_2) = w_1 f(x_1) + w_2 f(x_2) = Q_2(f).$$

Therefore,

$$E_2(f) := I(f) - Q_2(f) = \int_0^\infty \big(f(t) - H_3(t)\big)e^{-t}\,dt.$$

Then by Hermite remainder:

$$E_2(f) = \int_0^\infty \frac{f^{(4)}(\xi_t)}{4!}\,(t-x_1)^2(t-x_2)^2\,e^{-t}\,dt.$$

Note that $(t-x_1)^2(t-x_2)^2e^{-t} \geq 0$ on $[0,\infty)$. Since $f^{(4)}$ is continuous, the integral mean value theorem implies there exists $\tau > 0$ such that

$$\boxed{E_2(f) = \frac{f^{(4)}(\tau)}{4!}\int_0^\infty (t-x_1)^2(t-x_2)^2e^{-t}\,dt = \frac{f^{(4)}(\tau)}{6} \qquad \tau > 0.}$$

(c)

Let $f(t) = \dfrac{1}{1+t}$. Then

$$I \approx w_1 f(t_1) + w_2 f(t_2) = \frac{2+\sqrt{2}}{4}\cdot\frac{1}{3-\sqrt{2}} + \frac{2-\sqrt{2}}{4}\cdot\frac{1}{3+\sqrt{2}} = \frac{4}{7} \approx 0.5714285714.$$

Compute

$$f^{(4)}(t) = \frac{4!}{(1+t)^5} = \frac{24}{(1+t)^5},$$

the remainder

$$E_2(f) = \frac{1}{6}f^{(4)}(\tau) = \frac{1}{6}\cdot\frac{24}{(1+\tau)^5} = \boxed{\frac{4}{(1+\tau)^5}}, \qquad \tau > 0.$$

In particular, since $f^{(4)}(t) \leq f^{(4)}(0) = 24$ for $t \geq 0$,

$$0 < E_2(f) \leq \frac{1}{6}\cdot 24 = 4,$$

True error and identification of τ. Using the given exact value $I = 0.596347361\ldots$,

$$\text{true error } = I - Q_2(f) = 0.596347361\ldots - \frac{4}{7} = 0.02491878957\ldots$$

Since $E_2(f) = \dfrac{4}{(1+\tau)^5}$, we can identify τ by

$$\frac{4}{(1+\tau)^5} = I - \frac{4}{7} \qquad \Rightarrow \qquad \tau = \left(\frac{4}{I-\frac{4}{7}}\right)^{1/5} - 1 \approx 1.761255601.$$

IV. Remainder of Gauss formulas

(a)

We know

$$\ell_m(t) = \prod_{\substack{j=1\\ j\neq m}}^{n} \frac{t-x_j}{x_m-x_j}, \qquad m = 1,\ldots,n.$$

Then

$$\ell_m(x_k) = \delta_{mk}.$$

$$p(t) = \sum_{m=1}^{n}\left[h_m(t)f_m + q_m(t)f'_m\right]$$

satisfies

$$p(x_k) = f_k, \qquad p'(x_k) = f'_k, \qquad k = 1,\ldots,n.$$

It is sufficient to impose the cardinal conditions

$$h_m(x_k) = \delta_{mk}, \quad h'_m(x_k) = 0, \qquad q_m(x_k) = 0, \quad q'_m(x_k) = \delta_{mk} \qquad (k = 1,\ldots,n).$$

Take

$$q_m(t) = (t - x_m)\,\ell_m(t)^2.$$

Then for every k,

$$q_m(x_k) = (x_k - x_m)\,\ell_m(x_k)^2 = 0,$$

and for $k \neq m$ we also have $q_m'(x_k) = 0$ because $\ell_m(x_k) = 0$. At $k = m$,

$$q_m'(t) = \ell_m(t)^2 + (t - x_m) \cdot 2\ell_m(t)\ell_m'(t) \quad \Rightarrow \quad q_m'(x_m) = \ell_m(x_m)^2 = 1.$$

Hence q_m satisfies the required conditions.

Moreover,

$$q_m(t) = (t - x_m)\ell_m(t)^2 = (c_m + d_m t)\ell_m(t)^2$$

with

$$\boxed{c_m = -x_m, \qquad d_m = 1.}$$

Consider

$$h_m(t) = \big(A_m + B_m(t - x_m)\big)\ell_m(t)^2.$$

Then $h_m(x_k) = 0$ for $k \neq m$, and $h_m(x_m) = A_m$. To enforce $h_m(x_m) = 1$, take $A_m = 1$.

Differentiate:

$$h_m'(t) = B_m\ell_m(t)^2 + \big(1 + B_m(t - x_m)\big) \cdot 2\ell_m(t)\ell_m'(t).$$

For $k \neq m$, $\ell_m(x_k) = 0$ implies $h_m'(x_k) = 0$ automatically. At $k = m$,

$$h_m'(x_m) = B_m\ell_m(x_m)^2 + 1 \cdot 2\ell_m(x_m)\ell_m'(x_m) = B_m + 2\ell_m'(x_m).$$

Imposing $h_m'(x_m) = 0$ gives

$$B_m = -2\ell_m'(x_m).$$

Therefore

$$h_m(t) = \Big(1 - 2(t - x_m)\ell_m'(x_m)\Big)\ell_m(t)^2.$$

Finally, rewriting in the requested form $h_m(t) = (a_m + b_m t)\ell_m(t)^2$,

$$1 - 2(t - x_m)\ell_m'(x_m) = 1 + 2x_m\ell_m'(x_m) - 2\ell_m'(x_m)\,t,$$

so

$$\boxed{a_m = 1 + 2x_m\ell_m'(x_m), \qquad b_m = -2\ell_m'(x_m).}$$

$$\boxed{h_m(t) = \Big(1 - 2(t - x_m)\ell_m'(x_m)\Big)\ell_m(t)^2, \qquad q_m(t) = (t - x_m)\ell_m(t)^2,}$$

with

$$\boxed{a_m = 1 + 2x_m\ell_m'(x_m), \quad b_m = -2\ell_m'(x_m), \quad c_m = -x_m, \quad d_m = 1.}$$

(b)

Let

$$I(f) := \int_a^b f(t)\,\rho(t)\,dt$$

be the weighted integral functional (with a given weight ρ). From (a), there exist Hermite cardinal basis functions h_m, q_m satisfying

$$p(t) = \sum_{m=1}^{n} \big[h_m(t) f_m + q_m(t) f_m'\big].$$

Define

$$w_k := I(h_k) = \int_a^b h_k(t)\rho(t)\,dt, \qquad \mu_k := I(q_k) = \int_a^b q_k(t)\rho(t)\,dt, \qquad k = 1, \dots, n.$$

Then

$$I(p) = \sum_{m=1}^{n} \Big[I(h_m) f_m + I(q_m) f_m'\Big] = \sum_{m=1}^{n} \big[w_m f_m + \mu_m f_m'\big].$$

For any polynomial $p \in \mathbb{P}_{2n-1}$, choose $f_m = p(x_m)$ and $f'_m = p'(x_m)$. Then the Hermite interpolant produced by before is exactly p itself (uniqueness of Hermite interpolation in $\mathbb{P}_{2n-1}$). Hence

$$I(p) = \sum_{k=1}^{n} \big[w_k p(x_k) + \mu_k p'(x_k)\big] = I_n(p),$$

so the error functional

$$E_n(p) = I(p) - I_n(p)$$

satisfies

$$\boxed{E_n(p) = 0 \quad \text{for all } p \in \mathbb{P}_{2n-1}.}$$

$$\boxed{I_n(f) = \sum_{k=1}^{n} \big[w_k f(x_k) + \mu_k f'(x_k)\big], \qquad w_k = \int_a^b h_k(t)\rho(t)\,dt, \quad \mu_k = \int_a^b q_k(t)\rho(t)\,dt.}$$

(c)

$$\mu_k = I(q_k) = \int_a^b q_k(t)\rho(t)\,dt, \qquad q_k(t) = (t - x_k)\,\ell_k(t)^2,$$

where ℓ_k is the Lagrange basis at the nodes $x_1, \dots, x_n$. Let the node (Vandermonde) polynomial be

$$\omega_n(t) = \prod_{j=1}^{n} (t - x_j).$$

Since $\ell_k(t) = \dfrac{\omega_n(t)}{(t - x_k)\omega'_n(x_k)}$, we can rewrite q_k as

$$q_k(t) = (t - x_k)\left(\frac{\omega_n(t)}{(t - x_k)\omega'_n(x_k)}\right)^2 = \frac{\omega_n(t)^2}{(t - x_k)\,\omega'_n(x_k)^2}.$$

Hence

$$\mu_k = \frac{1}{\omega'_n(x_k)^2} \int_a^b \frac{\omega_n(t)^2}{t - x_k}\,\rho(t)\,dt.$$

Therefore, $\mu_k = 0$ for every $k = 1, \dots, n$ if and only if

$$\int_a^b \frac{\omega_n(t)^2}{t - x_k}\,\rho(t)\,dt = \int_a^b \omega_n(t)\ell_k(t)\rho(t)\,dt = 0, \qquad k = 1, \dots, n.$$

since $\omega'_n(x_k) \neq 0$ for distinct nodes. Now for any $q \in \mathbb{P}_{n-1}$ we have the Lagrange representation

$$q(t) = \sum_{k=1}^{n} q(x_k)\ell_k(t).$$

Therefore,

$$\int_a^b q(t)\omega_n(t)\rho(t)\,dt = \sum_{k=1}^{n} q(x_k) \int_a^b \omega_n(t)\ell_k(t)\rho(t)\,dt = 0,$$

which shows that $\omega_n \perp \mathbb{P}_{n-1}$ with respect to the weight ρ. Hence ω_n is (up to a constant factor) the degree-n orthogonal polynomial for ρ, and the nodes x_k are its zeros.

V. Prove a lemma

(a)

By Taylor's expansion with Lagrange remainder with $u \in \mathcal{C}^4(\mathbb{R})$:

$$u(\bar{x} - h) - 2u(\bar{x}) + u(\bar{x} + h) = h^2 u''(\bar{x}) + \frac{-h^3}{3!}u'''(\bar{x}) + \frac{h^3}{3!}u'''(\bar{x}) + 2\frac{h^4}{4!}u^{(4)}(\xi(\bar{x}))$$

So,

$$RHS = u''(\bar{x}) + O(h^2)$$

which is second-accurate. Then with the random errors:

$$|u''(\bar{x}) - D^2u(\bar{x})| = |\frac{\varepsilon_1 + 2\varepsilon_2 + \varepsilon_3}{h^2} + 2\frac{h^2}{4!}u^{(4)}(\xi(\bar{x}))| \leqslant \frac{4E}{h^2} + \frac{h^2}{12}|u^{(4)}(\xi(\bar{x}))|$$

And since the $\xi(\bar{x})$ is from the taylor expansion, So $\xi(\bar{x}) \in (\bar{x}-h, \bar{x}+h)$.
Assume $M = \max_{x\in(\bar{x}-h,\bar{x}+h)} |u^{(4)}(x)|$, Then

$$|u''(\bar{x}) - D^2u(\bar{x})| \leqslant \frac{4E}{h^2} + \frac{h^2}{12}M, \quad \forall h$$

Then by basic equality RHS get the minimal value when $h = (48\frac{E}{M})^{\frac{1}{4}}$. RHS $= 2\sqrt{\frac{ME}{3}}$
We suppuse

$$D_4^2u(\bar{x}) = \frac{a\,u(\bar{x}-2h) + b\,u(\bar{x}-h) + c\,u(\bar{x}) + b\,u(\bar{x}+h) + a\,u(\bar{x}+2h)}{h^2}.$$

Taylor expansions about $\bar{x}$:

$$u(\bar{x}\pm h) = u \pm hu' + \frac{h^2}{2}u'' \pm \frac{h^3}{6}u^{(3)} + \frac{h^4}{24}u^{(4)} \pm \frac{h^5}{120}u^{(5)} + \frac{h^6}{720}u^{(6)} + O(h^7),$$

$$u(\bar{x}\pm 2h) = u \pm 2hu' + \frac{(2h)^2}{2}u'' \pm \frac{(2h)^3}{6}u^{(3)} + \frac{(2h)^4}{24}u^{(4)} \pm \frac{(2h)^5}{120}u^{(5)} + \frac{(2h)^6}{720}u^{(6)} + O(h^7).$$

By symmetry, all odd-derivative terms cancel. Matching coefficients gives

$$2a + 2b + c = 0, \qquad 4a + b = 1, \qquad 16a + b = 0,$$

hence

$$a = -\frac{1}{12}, \qquad b = \frac{4}{3}, \qquad c = -\frac{5}{2}.$$

Therefore the 4th-order accurate symmetric formula is

$$D_4^2u(\bar{x}) = \frac{-u(\bar{x}-2h) + 16u(\bar{x}-h) - 30u(\bar{x}) + 16u(\bar{x}+h) - u(\bar{x}+2h)}{12h^2}.$$

Moreover, for some $\xi \in (\bar{x}-2h, \bar{x}+2h)$,

$$u''(\bar{x}) - D_4^2u(\bar{x}) = -\frac{h^4}{90}\,u^{(6)}(\xi),$$

so

$$\left|u''(\bar{x}) - D_4^2u(\bar{x})\right| \leq \frac{h^4}{90}\max_{x\in(\bar{x}-2h,\bar{x}+2h)} |u^{(6)}(x)|.$$

$$\left|u''(\bar{x}) - D_4^2u(\bar{x})\right| \leq \frac{M_6}{90}h^4 + \frac{16E}{3h^2}, \qquad M_6 := \max_{x\in(\bar{x}-2h,\bar{x}+2h)} |u^{(6)}(x)|.$$

Minimizing the upper bound yields

$$h_4 = \left(\frac{240E}{M_6}\right)^{1/6}, \qquad \Phi_4(h_4) = \left(\frac{32}{15}\right)^{1/3} M_6^{1/3}E^{2/3}.$$

- **Optimal step size.** Since $h_2 \propto E^{1/4}$ while $h_4 \propto E^{1/6}$, for small noise levels $E \in (0,1)$ we have $E^{1/6} > E^{1/4}$, hence

  $$h_4 \text{ is typically larger than } h_2.$$

  In other words, the 4th-order method does not require taking h extremely small; making h too small mainly amplifies noise.

- **Minimized error bound vs noise level.** The optimized bounds scale as

  $$\Phi_2(h_2) \propto E^{1/2}, \qquad \Phi_4(h_4) \propto E^{2/3}.$$

  For $E \in (0,1)$, $E^{2/3} < E^{1/2}$, so the 4th-order stencil yields a smaller optimal upper bound (a better truncation–noise tradeoff).

- **Cost/regularity.** The 4th-order stencil uses more points (5 instead of 3) and requires higher smoothness (involving $u^{(6)}$). Its coefficients are larger in magnitude, which increases the noise-amplification constant in the E/h^2 term, but after optimizing h the faster h^4 truncation decay typically dominates.